

File permissions in Linux

Project description

Changing Permissions of files and directories for organization to update security and ensure least privilege.

Check file and directory details

Used the command `ls -la` to display the permissions of all files and directories including hidden files.

```
researcher2@312cb49bfe33:~$ cd projects
researcher2@312cb49bfe33:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 13 12:56 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 13 13:56 ..
-rw--w---- 1 researcher2 research_team  46 Sep 13 12:56 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 13 12:56 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Sep 13 12:56 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep 13 12:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 13 12:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 13 12:56 project_t.txt
researcher2@312cb49bfe33:~/projects$
```

Describe the permissions string

drwxrwxrwx

d- indicates it is a directory

- - indicates it is a file

The following three groups of rwx describes permissions for users, groups, and other. r indicating read permissions, w indicating write permissions, and x indicating executable permissions include viewing directories.

Change file permissions

The organization does not allow others to have write access to any files therefore the file project_k.txt needs to have that permission removed.

`chmod o-w project_k.txt` was used to take away write permissions from other.

```
researcher2@312cb49bfe33:~$ cd projects
researcher2@312cb49bfe33:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 13 12:56 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 13 13:56 ..
-rw--w---- 1 researcher2 research_team  46 Sep 13 12:56 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 13 12:56 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Sep 13 12:56 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Sep 13 12:56 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 13 12:56 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 13 12:56 project_t.txt
researcher2@312cb49bfe33:~/projects$ chmod o-w project_k.txt
researcher2@312cb49bfe33:~/projects$
```

Change file permissions on a hidden file

Permissions were changed so that the user and group could read but not write the hidden file using

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`chmod u-w,g-w,g+r .project_x.txt`

```
researcher2@a61be5ae321a:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 13 13:48 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 13 14:46 ..
-rw--w---- 1 researcher2 research_team  46 Sep 13 13:48 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 13 13:48 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep 13 13:48 project_k.txt
-rw----- 1 researcher2 research_team  46 Sep 13 13:48 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 13 13:48 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 13 13:48 project_t.txt
researcher2@a61be5ae321a:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@a61be5ae321a:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Sep 13 13:48 .
drwxr-xr-x 3 researcher2 research_team 4096 Sep 13 14:46 ..
-r--r----- 1 researcher2 research_team  46 Sep 13 13:48 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Sep 13 13:48 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Sep 13 13:48 project_k.txt
-rw----- 1 researcher2 research_team  46 Sep 13 13:48 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 13 13:48 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Sep 13 13:48 project_t.txt
researcher2@a61be5ae321a:~/projects$
```

Change directory permissions

The drafts directory and its contents should only be accessible by researcher 2 so the permissions needed to be modified.

```
researcher2@a61be5ae321a:~/projects$ cd /home/researcher2/projects  
researcher2@a61be5ae321a:~/projects$ chmod g-x drafts
```

Summary

Using the command `chmod`, permissions were taken away or given for certain files and directories. The command `ls -la` was used to determine which files had which permissions and if the command to change permissions took effect. This command also displayed hidden files which allowed the ability to view what permissions that hidden file had as well as modify its permissions to strive for least privilege within organization.